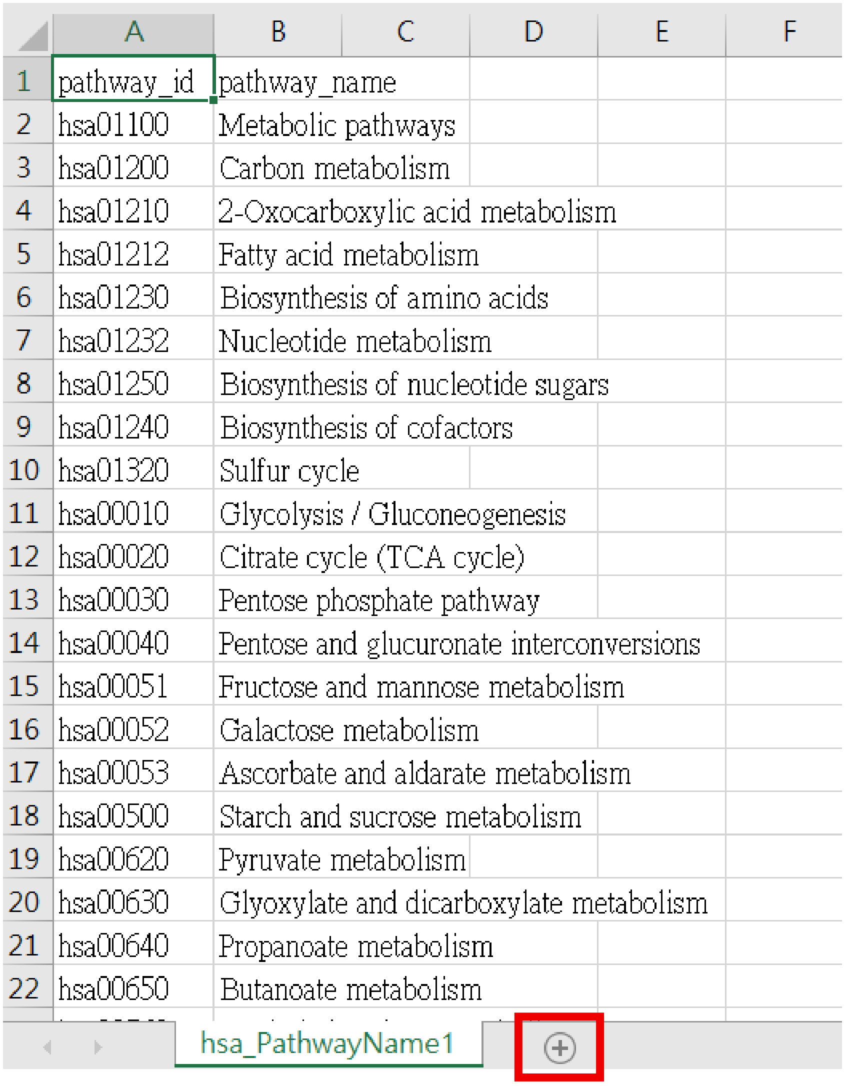
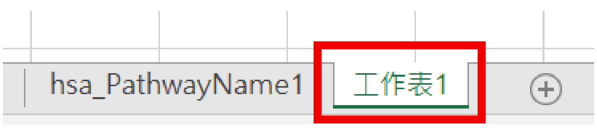
資料表建置教學

1. 將.xlsx 檔轉存成.csv 檔
2.



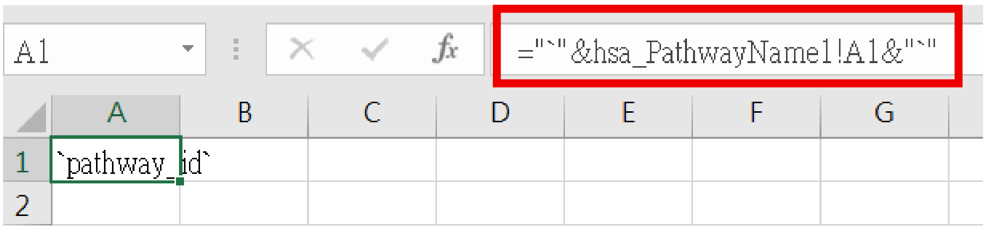
	A	B
1	pathway_id	pathway_name
2	hsa01100	Metabolic pathways
3	hsa01200	Carbon metabolism
4	hsa01210	2-Oxocarboxylic acid metabolism
5	hsa01212	Fatty acid metabolism
6	hsa01230	Biosynthesis of amino acids
7	hsa01232	Nucleotide metabolism
8	hsa01250	Biosynthesis of nucleotide sugars
9	hsa01240	Biosynthesis of cofactors
10	hsa01320	Sulfur cycle
11	hsa00010	Glycolysis / Gluconeogenesis
12	hsa00020	Citrate cycle (TCA cycle)
13	hsa00030	Pentose phosphate pathway
14	hsa00040	Pentose and glucuronate interconversions
15	hsa00051	Fructose and mannose metabolism
16	hsa00052	Galactose metabolism
17	hsa00053	Ascorbate and aldarate metabolism
18	hsa00500	Starch and sucrose metabolism
19	hsa00620	Pyruvate metabolism
20	hsa00630	Glyoxylate and dicarboxylate metabolism
21	hsa00640	Propanoate metabolism
22	hsa00650	Butanoate metabolism

開新的工作頁

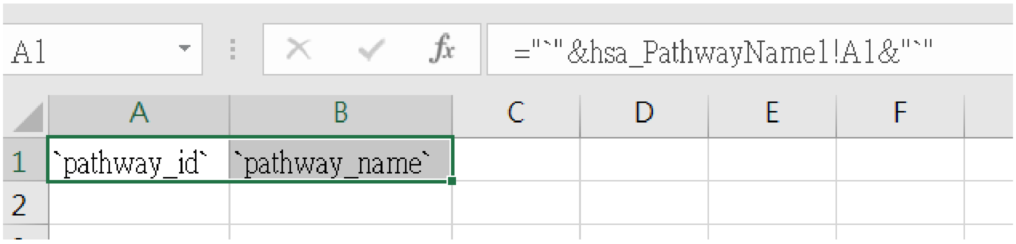


使用 excel 公式：

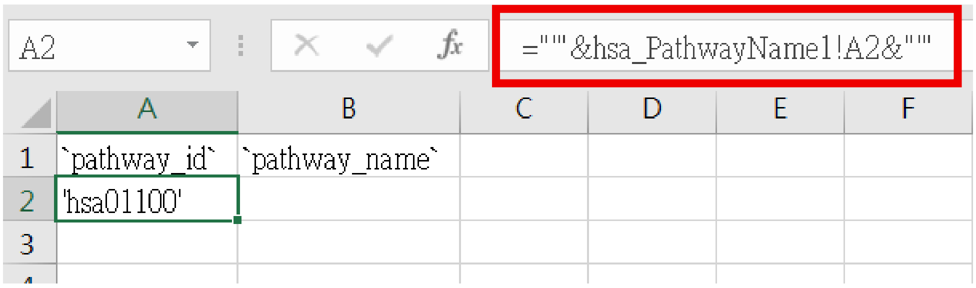
欄位名稱：=" `"&對應欄位&" `"



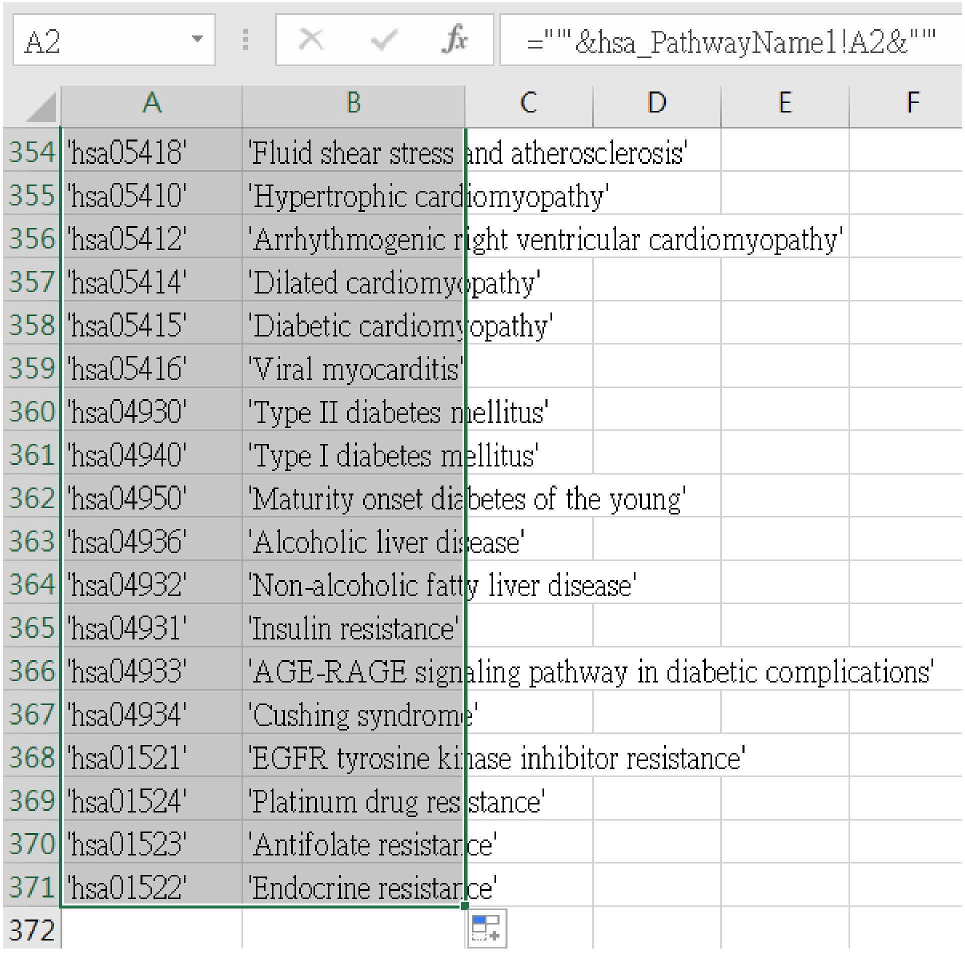
有幾個欄位就將已寫好的公式往右方拉到對應位置(在欄位右下方會出現+的符號，滑鼠按住往右方拖曳)



剩下列的值：=" `"&對應欄位&" `"



有幾個值就將已寫好的公式往右方拉到對應位置(在欄位右下方會出現+的符號，滑鼠按住往右方拖曳)，往右拉完再往下拉到底部



3. 將此新增的工作分頁的內容全選並複製(ctrl+A、ctrl+C)，到原本的工作分頁的第一欄第一列滑鼠按右鍵，選擇"只貼上值"取代原本的內容，再將新增的工作分頁刪除，並存檔
4. 打開文字編輯器(推薦使用 notepad++)

5. 先建立空的資料表

在文字編輯器中先打好要執行的 SQL 指令

```
CREATE TABLE `kegg_hsa_metabolism_pathways` (
	`pathway_id` VARCHAR(50) NOT NULL COLLATE 'utf8mb4_0900_ai_ci',
	`metabolite_type` VARCHAR(50) NOT NULL COLLATE 'utf8mb4_0900_ai_ci',
	`kegg_id` VARCHAR(50) NOT NULL COLLATE 'utf8mb4_0900_ai_ci',
	`metabolite_name` VARCHAR(400) NULL DEFAULT NULL COLLATE 'utf8mb4_0900_ai_ci'
)
COLLATE='utf8mb4_0900_ai_ci'
ENGINE=InnoDB
;
```

通用建立空的資料表指令，黃色空格要改成要建的資料表的內容：

CREATE TABLE `資料表名稱` (

`欄位名稱` **VARCHAR**(50) **NOT NULL COLLATE** 'utf8mb4_0900_ai_ci',

`欄位名稱` **DOUBLE NULL DEFAULT NULL,**

`欄位名稱` **VARCHAR**(400) **NULL DEFAULT NULL COLLATE** 'utf8mb4_0900_ai_ci'

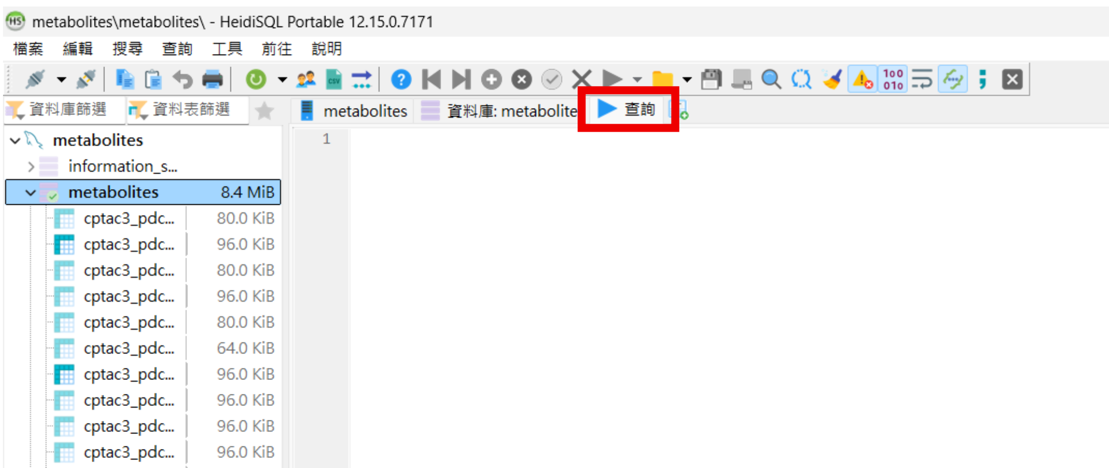
)

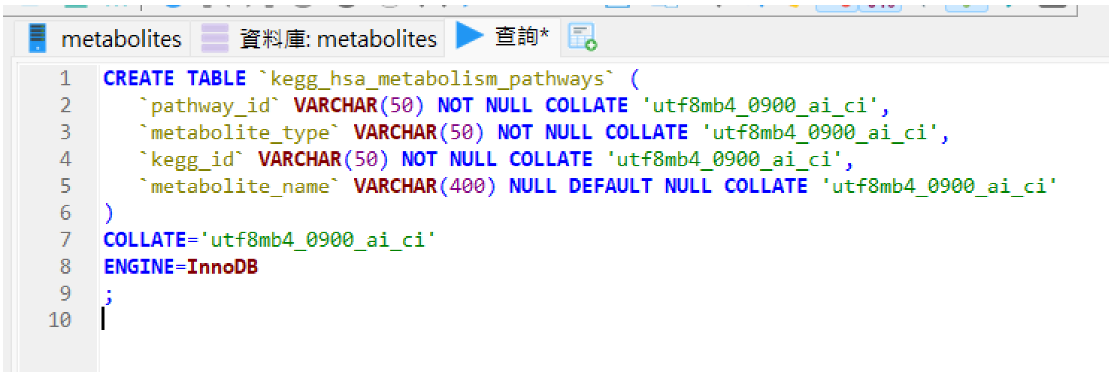
COLLATE='utf8mb4_0900_ai_ci'

ENGINE=InnoDB

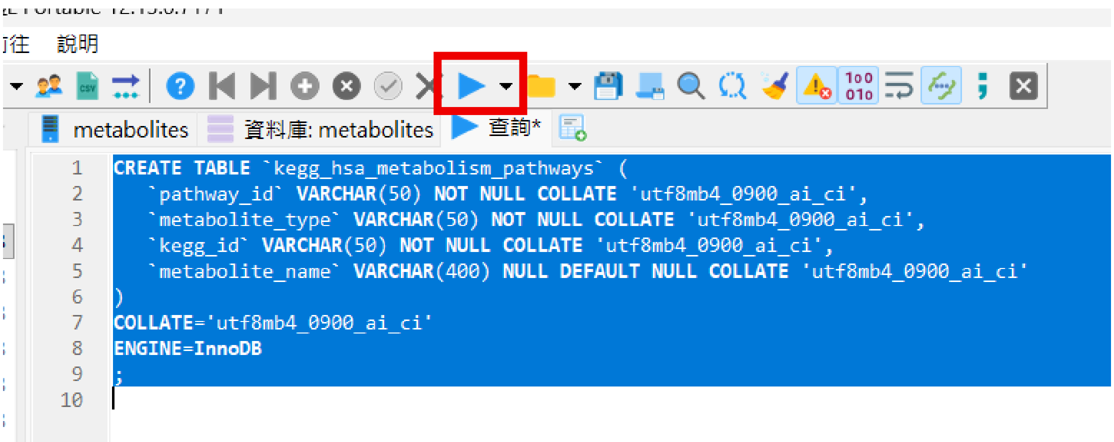
;

6. 複製指令並在 HeidiSQL 中的查詢貼上





執行：全選(ctrl+A)並點選執行



指令執行完後即可把內容刪除，不須存檔，若是想保留指令也可存檔

7. 在文字編輯器中打開 csv 檔(會看到資料用逗號做連接)

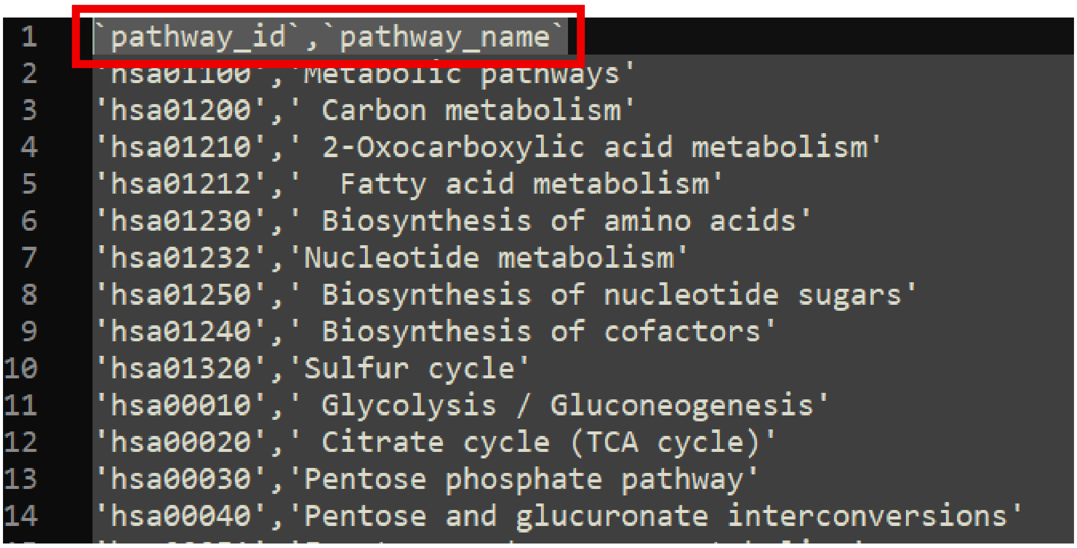
```
`pathway_id`,`pathway_name`
'hsa01100','Metabolic pathways'
'hsa01200',' Carbon metabolism'
'hsa01210',' 2-Oxocarboxylic acid metabolism'
'hsa01212','  Fatty acid metabolism'
'hsa01230',' Biosynthesis of amino acids'
'hsa01232','Nucleotide metabolism'
'hsa01250',' Biosynthesis of nucleotide sugars'
'hsa01240',' Biosynthesis of cofactors'
'hsa01320','Sulfur cycle'
'hsa00010',' Glycolysis / Gluconeogenesis'
'hsa00020',' Citrate cycle (TCA cycle)'
'hsa00030','Pentose phosphate pathway'
'hsa00040','Pentose and glucuronate interconversions'
'hsa00051','Fructose and mannose metabolism'
'hsa00052','Galactose metabolism'
'hsa00053',' Ascorbate and aldarate metabolism'
'hsa00500',' Starch and sucrose metabolism'
'hsa00620','Pyruvate metabolism'
'hsa00630',' Glyoxylate and dicarboxylate metabolism'
'hsa00640',' Propanoate metabolism'
'hsa00650',' Butanoate metabolism'
'hsa00562',' Inositol phosphate metabolism'
'hsa00190',' Oxidative phosphorylation'
'hsa00910','  Nitrogen metabolism'
'hsa00920',' Sulfur metabolism'
'hsa00061',' Fatty acid biosynthesis'
'hsa00062',' Fatty acid elongation'
'hsa00071',' Fatty acid degradation'
'hsa00100','Steroid biosynthesis'
'hsa00120',' Primary bile acid biosynthesis'
'hsa00140',' Steroid hormone biosynthesis'
'hsa00561',' Glycerolipid metabolism'
'hsa00564',' Glycerophospholipid metabolism'
```

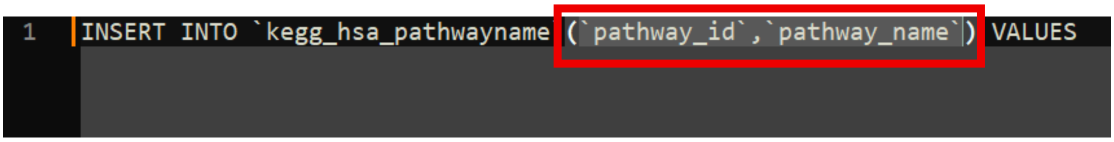
8. 開新的文字編輯頁面

9. 在新的文字編輯頁面中，貼上 Insert 指令(有幾個欄位就寫幾個欄位名稱)

INSERT INTO `資料表名稱`(`欄位名稱`,`欄位名稱`) VALUES

欄位名稱可直接到已在文字編輯器中打開的 csv 檔中剪下(ctrl+X)

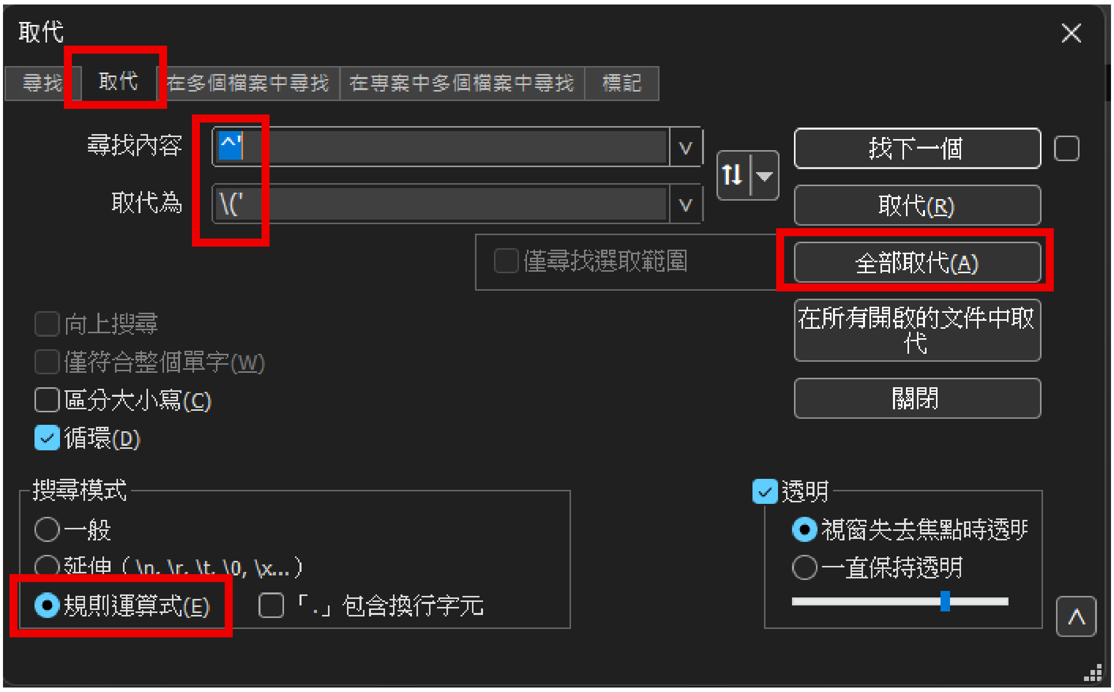




10. 到在文字編輯器中開啟的 csv 檔進行搜尋與取代，一直到使用完，此檔案都不須存檔，僅做搜尋與取代的動作方便建立資料表的 Insert 指令

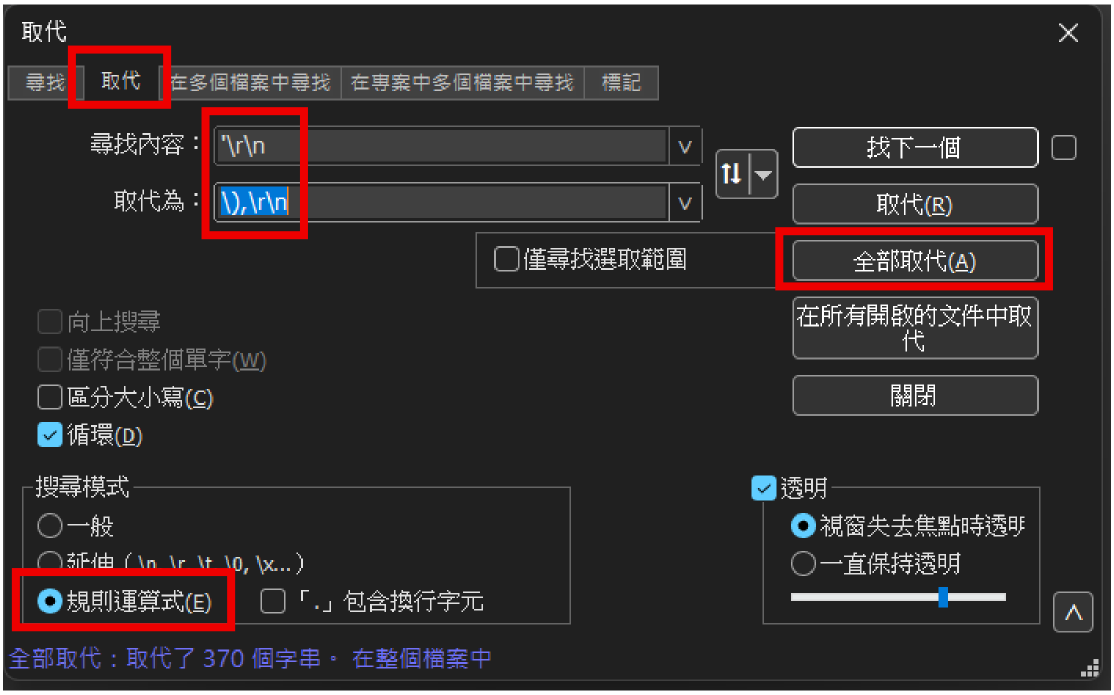
尋找內容：^'

取代為：\('

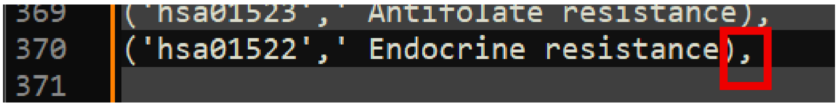


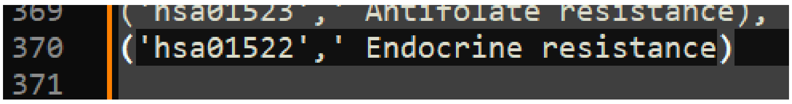
尋找內容：'\r\n

取代為：\),\r\n



往下拉找到最後一列，將最後一個逗號刪除





11. 將取代完後的內容全選並複製(ctrl+A、ctrl+C)，貼到 Insert 指令的下方

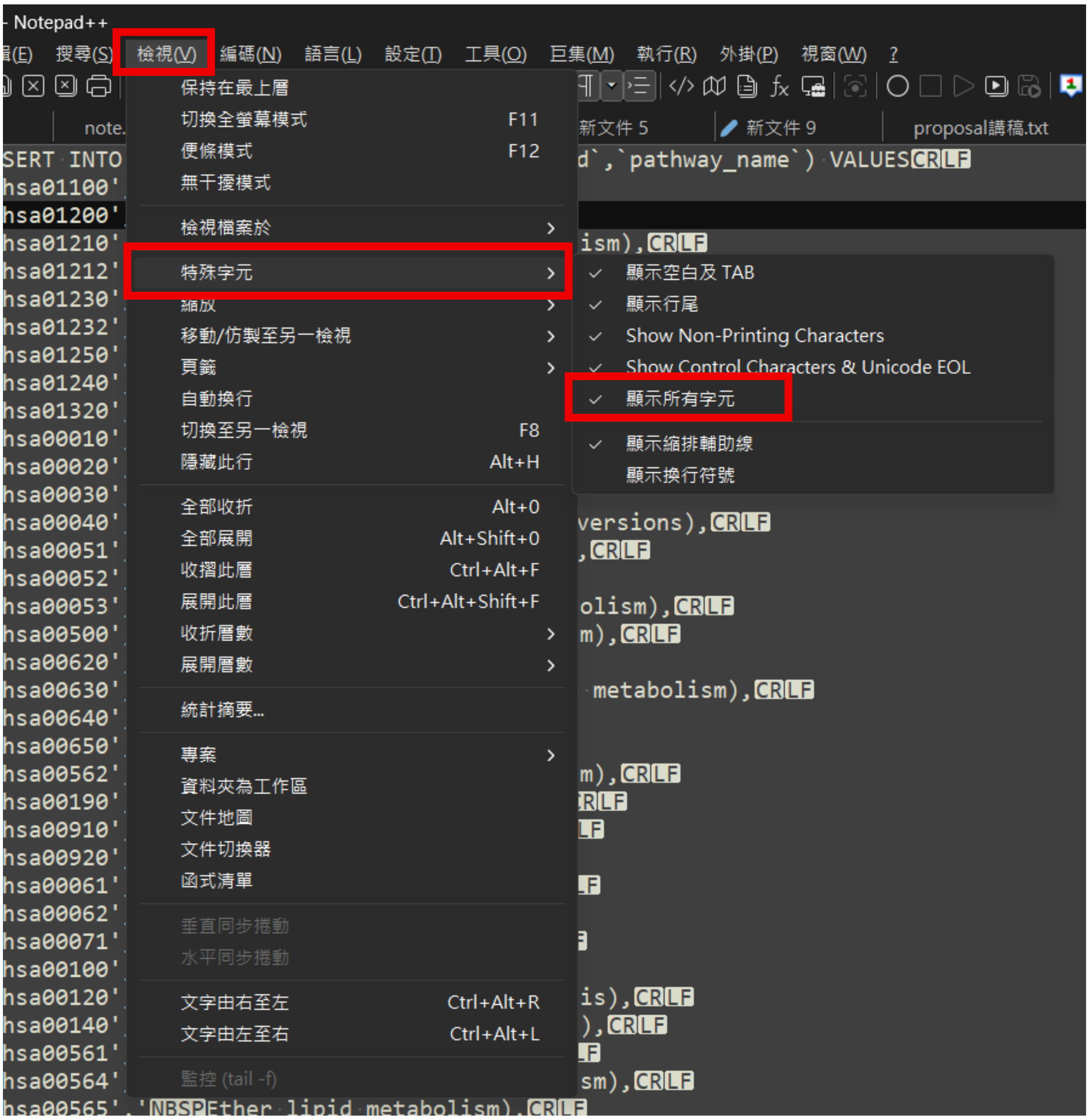
```
INSERT INTO `kegg_hsa_pathwayname`(`pathway_id`,`pathway_name`) VALUES
('hsa01100','Metabolic pathways),
('hsa01200',' Carbon metabolism),
('hsa01210',' 2-Oxocarboxylic acid metabolism),
('hsa01212','  Fatty acid metabolism),
('hsa01230',' Biosynthesis of amino acids),
('hsa01232','Nucleotide metabolism),
('hsa01250',' Biosynthesis of nucleotide sugars),
('hsa01240',' Biosynthesis of cofactors),
('hsa01320','Sulfur cycle),
('hsa00010',' Glycolysis / Gluconeogenesis),
('hsa00020',' Citrate cycle (TCA cycle)),
('hsa00030','Pentose phosphate pathway),
('hsa00040','Pentose and glucuronate interconversions),
('hsa00051','Fructose and mannose metabolism),
('hsa00052','Galactose metabolism),
('hsa00053',' Ascorbate and aldarate metabolism),
('hsa00500',' Starch and sucrose metabolism),
('hsa00620','Pyruvate metabolism),
('hsa00630',' Glyoxylate and dicarboxylate metabolism),
('hsa00640',' Propanoate metabolism),
('hsa00650',' Butanoate metabolism),
('hsa00562',' Inositol phosphate metabolism),
('hsa00190',' Oxidative phosphorylation),
```

12. 將此頁內容全選並複製(ctrl+A、ctrl+C)，貼到 HeidiSQL 中的查詢並執行

```
INSERT INTO `kegg_hsa_pathwayname`(`pathway_id`,`pathway_name`) VALUES
('hsa01100','Metabolic pathways),
('hsa01200','Carbon metabolism),
('hsa01210','2-Oxocarboxylic acid metabolism),
('hsa01212','Fatty acid metabolism),
('hsa01230','Biosynthesis of amino acids),
('hsa01232','Nucleotide metabolism),
('hsa01250','Biosynthesis of nucleotide sugars),
('hsa01240','Biosynthesis of cofactors),
('hsa01320','Sulfur cycle),
('hsa00010','Glycolysis / Gluconeogenesis),
('hsa00020','Citrate cycle (TCA cycle)),
('hsa00030','Pentose phosphate pathway),
('hsa00040','Pentose and glucuronate interconversions),
('hsa00051','Fructose and mannose metabolism),
('hsa00052','Galactose metabolism),
('hsa00053','Ascorbate and aldarate metabolism),
('hsa00500','Starch and sucrose metabolism),
('hsa00620','Pyruvate metabolism),
('hsa00630','Glyoxylate and dicarboxylate metabolism),
('hsa00640','Propanoate metabolism),
('hsa00650','Butanoate metabolism),
('hsa00562','Inositol phosphate metabolism),
('hsa00190','Oxidative phosphorylation),
('hsa00910','Nitrogen metabolism),
('hsa00920','Sulfur metabolism),
('hsa00061','Fatty acid biosynthesis),
```

在前面文字編輯器處理 csv 檔時，有時會有雙引號出現，記得將雙引號取代為空白；有時候會有特殊字元或多餘的空白，也可以使用取代功能一次性取代為空白；還有一點也很重要，若是在資料的值中名稱有單引號'，要 csv 檔剛開起的時候先利用取代功能將單引號'取代成\'，因為在 SQL 語法中，會影響到資料所在的位置以及值會跑掉。

在使用 notpad++時，我個人習慣將所有字元都顯示出來，雖然在閱讀上會有點亂，但是在要取代特殊字元時，很明顯可以看到特殊字元，就不容易漏掉。



特殊字元舉例：NBSP 在沒有顯示特殊字元的情況下只是個空白